In [2]:
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"""
Experiment: Adversarial Attacks, Model Poisoning, Evasion Techniques,
and Ethical Challenges in ML-Based Cyber Defense
Dataset:
    adversarial_cyber_dataset.csv
Label meaning:
    0 = Normal traffic
    1 = Attack traffic
This program demonstrates:
1. Training a baseline ML cyber defense model.
2. Evasion attack: modifying malicious samples at test time.
3. Model poisoning: corrupting training labels.
4. Ethical challenges in ML-based cyber defense.
"""
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)
# ------------------------------------------------
# 1. Load dataset
# ------------------------------------------------
df = pd.read_csv("adversarial_cyber_dataset.csv")
print("First five records of dataset:")
print(df.head())
print("\nDataset shape:", df.shape)
print("\nClass distribution:")
print(df["label"].value_counts())
# ------------------------------------------------
# 2. Separate features and label
# ------------------------------------------------
X = df.drop("label", axis=1)
y = df["label"]
feature_names = X.columns.tolist()
# ------------------------------------------------
# 3. Split dataset into training and testing parts
# ------------------------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=42,
    stratify=y
)
# ------------------------------------------------
# 4. Normalize numerical features
# ------------------------------------------------
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
# ------------------------------------------------
# 5. Train baseline cyber defense model
# ------------------------------------------------
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
model.fit(X_train_scaled, y_train)
baseline_predictions = model.predict(X_test_scaled)
print("\n================ BASELINE MODEL ================")
print("Accuracy:", accuracy_score(y_test, baseline_predictions))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, baseline_predictions))
print("\nClassification Report:")
print(classification_report(y_test, baseline_predictions))
# ------------------------------------------------
# 6. Evasion attack simulation
# ------------------------------------------------
"""
Evasion attack means modifying malicious test samples
to make them resemble normal traffic.
This controlled simulation shifts numeric features
35% toward the average normal traffic profile.
"""
# FIX 1: Convert test features to float64
X_test_evasion = X_test.copy().astype(np.float64)
# FIX 2: Ensure the normal profile uses float64
normal_profile = (
    X_train.loc[y_train == 0]
    .mean()
    .astype(np.float64)
)
attack_indices = y_test[y_test == 1].index
for idx in attack_indices:
    X_test_evasion.loc[idx, feature_names] = (
        0.65 * X_test_evasion.loc[idx, feature_names]
        + 0.35 * normal_profile.loc[feature_names]
    )
X_test_evasion_scaled = scaler.transform(X_test_evasion)
evasion_predictions = model.predict(X_test_evasion_scaled)
print("\n================ EVASION ATTACK TEST ================")
print(
    "Accuracy after evasion:",
    accuracy_score(y_test, evasion_predictions)
)
print("\nConfusion Matrix after evasion:")
print(confusion_matrix(y_test, evasion_predictions))
print("\nClassification Report after evasion:")
print(classification_report(y_test, evasion_predictions))
# ------------------------------------------------
# 7. Model poisoning attack simulation
# ------------------------------------------------
"""
Model poisoning means corrupting training labels.
```

```
Here, 20% of attack samples in the training set
are incorrectly relabelled as normal traffic.
"""
y_train_poisoned = y_train.copy()
attack_train_indices = y_train_poisoned[
    y_train_poisoned == 1
].index
np.random.seed(42)
poison_count = int(0.20 * len(attack_train_indices))
poisoned_indices = np.random.choice(
    attack_train_indices,
    size=poison_count,
    replace=False
)
y_train_poisoned.loc[poisoned_indices] = 0
poisoned_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
poisoned_model.fit(X_train_scaled, y_train_poisoned)
poisoned_predictions = poisoned_model.predict(X_test_scaled)
print("\n================ MODEL POISONING TEST ================")
print(
    "Accuracy after poisoning:",
    accuracy_score(y_test, poisoned_predictions)
)
print("\nConfusion Matrix after poisoning:")
print(confusion_matrix(y_test, poisoned_predictions))
print("\nClassification Report after poisoning:")
print(classification_report(y_test, poisoned_predictions))
# ------------------------------------------------
# 8. Feature importance
# ------------------------------------------------
importance = pd.DataFrame({
    "feature": feature_names,
    "importance": model.feature_importances_
}).sort_values(
    by="importance",
    ascending=False
)
print("\n================ FEATURE IMPORTANCE ================")
print(importance)
# ------------------------------------------------
# 9. Ethical discussion points
# ------------------------------------------------
print("\n================ ETHICAL CHALLENGES ================")
print("1. Use adversarial ML experiments only in controlled lab environments.")
print("2. Do not use evasion knowledge to bypass real security systems.")
print("3. Protect training datasets from unauthorized modification.")
print("4. Avoid collecting personal or sensitive user data unnecessarily.")
print("5. Explainable and auditable ML models are important in cyber defense.")
print("6. Human verification is required before serious actions like blocking users
```

```
First five records of dataset:
    duration   src_bytes   dst_bytes  packet_count  failed_logins  \
0  33.467407  531.185935  612.114777     18.926939              1
1  58.950554  381.135641  578.758203     26.792740              0
2  36.068054  578.166950  548.425743     29.032372              0
3  39.106649  581.551730  617.982927     20.393454              0
4  38.064571  443.368176  579.344254     24.101376              1

   same_srv_rate  serror_rate  label
0       0.649723     0.002102      0
1       0.665855     0.108698      0
2       0.745508     0.083698      0
3       0.852694     0.126892      0
4       0.637041     0.097181      0

Dataset shape: (240, 8)

Class distribution:
label
0    120
1    120
Name: count, dtype: int64

================ BASELINE MODEL ================
Accuracy: 0.9861111111111112

Confusion Matrix:
[[36  0]
 [ 1 35]]

Classification Report:
              precision    recall  f1-score   support

           0       0.97      1.00      0.99        36
           1       1.00      0.97      0.99        36

    accuracy                           0.99        72
   macro avg       0.99      0.99      0.99        72
weighted avg       0.99      0.99      0.99        72


================ EVASION ATTACK TEST ================
Accuracy after evasion: 0.9722222222222222

Confusion Matrix after evasion:
[[36  0]
 [ 2 34]]

Classification Report after evasion:
              precision    recall  f1-score   support

           0       0.95      1.00      0.97        36
           1       1.00      0.94      0.97        36

    accuracy                           0.97        72
   macro avg       0.97      0.97      0.97        72
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```
weighted avg       0.97      0.97      0.97        72


================ MODEL POISONING TEST ================
Accuracy after poisoning: 0.9722222222222222

Confusion Matrix after poisoning:
[[36  0]
 [ 2 34]]

Classification Report after poisoning:
              precision    recall  f1-score   support

           0       0.95      1.00      0.97        36
           1       1.00      0.94      0.97        36

    accuracy                           0.97        72
   macro avg       0.97      0.97      0.97        72
weighted avg       0.97      0.97      0.97        72


================ FEATURE IMPORTANCE ================
         feature  importance
6    serror_rate    0.289404
5  same_srv_rate    0.252390
3   packet_count    0.209303
1      src_bytes    0.087823
0       duration    0.081094
2      dst_bytes    0.075125
4  failed_logins    0.004861

================ ETHICAL CHALLENGES ================
1. Use adversarial ML experiments only in controlled lab environments.
2. Do not use evasion knowledge to bypass real security systems.
3. Protect training datasets from unauthorized modification.
4. Avoid collecting personal or sensitive user data unnecessarily.
5. Explainable and auditable ML models are important in cyber defense.
6. Human verification is required before serious actions like blocking users.
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